

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
Department of Computer Science and Engineering
CO1 ASSESSMENT TOOL 2 – Agile Sprint Planning & User Story Workshop

Course Code:	CSA10 / CSA1063	Course Title:	Software Engineering
CO Assessed:	CO1	Assessment:	Assessment Tool 2 – Agile Sprint Planning & User Story Workshop
Weightage:	20%	Total Marks:	25
CO1 Statement:	<i>Apply advanced methodologies and agile project management techniques to manage software projects effectively</i>		
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Date:	08-10-2026	Duration:	60 minutes

Code of Conduct

I Koppala Navya(192511231) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: Koppala Navya

Annexure A – Agile Sprint Planning & User Story Workshop

Instructions to Students:

Each student is assigned an individual software project problem statement. Based on the given problem statement, perform an **Agile Sprint Planning and User Story Workshop** by applying Agile software engineering practices. Prepare a comprehensive sprint planning document covering the following aspects:

Based on the given problem statement, perform an **Agile Sprint Planning and User Story Workshop** by applying Agile software engineering practices. Prepare a comprehensive sprint planning document covering the following aspects:

- 1. Identify the Project Vision and Stakeholders**
 - Define the project's vision, objectives, and identify all key stakeholders involved.
- 2. Create Epics and User Stories**
 - Break down the requirements into Epics and formulate well-structured User Stories using the format:
As a <user>, I want <feature>, so that <benefit>.
- 3. Define Acceptance Criteria**
 - Write clear and measurable acceptance criteria for each selected user story using the Given–When–Then format or equivalent.
- 4. Prioritize the Product Backlog**
 - Organize and prioritize the product backlog based on business value, customer needs, and project dependencies.
- 5. Plan the Sprint Backlog**
 - Select high-priority user stories for the first sprint and divide them into actionable development tasks.
- 6. Estimate Effort Using Story Points**
 - Assign story points to each selected user story using an appropriate estimation technique (e.g., Fibonacci sequence or Planning Poker) and justify your estimates.
- 7. Present the Sprint Goal and Expected Deliverables**
 - Define a clear sprint goal and describe the expected sprint deliverables, including the features to be completed at the end of the sprint.

Recommend safe and efficient routes based on real-time and historical conditions.

1. Identify the Project Vision and Stakeholders

Define the project's vision, objectives, and identify all key stakeholders involved.

Project Vision

The vision of the **Predictive Analytics System for Early Detection of Land Acquisition Delays** is to develop a data-driven system that can identify the possibility of land acquisition delays at an early stage. The system analyzes factors such as documentation status, landowner responses, legal issues, compensation status, and approval progress to predict potential delays and provide early warnings to authorities.

Objectives

- Predict possible land acquisition delays before they become critical.
- Identify the major factors responsible for delays.
- Provide early alerts to responsible officials.
- Support data-driven decision-making.
- Improve project planning and monitoring.
- Reduce the time and cost associated with delayed land acquisition.

Key Stakeholders

Key Stakeholders

Stakeholder	Role
Government Authorities	Manage roads and traffic
Traffic Authorities	Monitor real-time traffic
Project Managers	Track route efficiency
Data Analysts	Analyze traffic history
System Administrators	Maintain the system
Drivers/Commuters	Use recommended routes
Emergency Services	Find safe, fast routes
Logistics Companies	Optimize delivery routes
Road Authorities	Manage road conditions
Infrastructure Agencies	Plan future road development

2. Create Epics and User Stories

Break down the requirements into Epics and formulate well-structured User Stories using the format:

As a <user>, I want <feature>, so that <benefit>.

Epic 1 – Land Acquisition Data Management

User Story 1:

As a land acquisition officer, I want to enter and update land acquisition details, so that the system maintains accurate and current project information.

User Story 2:

As an administrator, I want to validate uploaded land records, so that incorrect or incomplete information does not affect predictions.

Epic 2 – Delay Factor Analysis

User Story 3:

As a project manager, I want to view factors affecting land acquisition progress, so that I can identify the reasons for potential delays.

User Story 4:

As a data analyst, I want to analyze historical acquisition data, so that important patterns related to delays can be identified.

Epic 3 – Delay Prediction

User Story 5:

As a project manager, I want the system to predict the likelihood of land acquisition delays, so that I can take preventive action early.

User Story 6:

As an acquisition officer, I want to receive an early warning when a delay risk is high, so that I can prioritize the affected cases.

Epic 4 – Dashboard and Reporting

User Story 7:

As a project manager, I want to view acquisition progress and predicted risks on a dashboard, so that I can monitor projects easily.

User Story 8:

As a government official, I want to generate reports on predicted delays, so that I can support planning and decision-making.

3. Define Acceptance Criteria

Write clear and measurable acceptance criteria for each selected user story using the Given–When–Then format or equivalent.

User Story 1 – Enter Land Acquisition Details

Given the officer is logged into the system,
When the officer enters valid land acquisition details and submits the form,
Then the system should validate and store the information successfully.

User Story 3 – Analyze Delay Factors

Given acquisition records are available,
When the project manager selects a project,
Then the system should display the major factors affecting acquisition progress.

User Story 5 – Predict Delay

Given valid land acquisition data is available,
When the user requests a prediction,
Then the system should calculate the delay risk and display a result such as **Low, Medium, or High**.

User Story 6 – Early Warning

Given a project has a high predicted delay risk,
When the prediction is generated,
Then the system should display an early warning and identify the affected project/case.

User Story 7 – Dashboard

Given the user has access to the dashboard,
When the dashboard is opened,
Then it should display acquisition progress, risk levels, and pending cases.

4. Prioritize the Product Backlog

Organize and prioritize the product backlog based on business value, customer needs, and project dependencies.

The backlog is prioritized using **business value, user importance, dependencies, and implementation feasibility**.

Priority	User Story	Business Value	Reason
1	Enter and update land data	Very High	Required for all other functions
2	Validate land records	Very High	Ensures reliable input data
3	Analyze historical data	High	Required for prediction development
4	Identify delay factors	High	Helps understand causes of delays
5	Predict delay risk	Very High	Core feature of the system
6	Generate early warnings	Very High	Enables preventive action
7	Dashboard	High	Provides easy monitoring
8	Reports	Medium	Supports documentation and decision-making

5. Plan the Sprint Backlog

Select high-priority user stories for the first sprint and divide them into actionable development tasks.

Sprint 1 – Data Management and Basic Prediction Preparation

For the first sprint, the team selects the highest-priority stories that form the foundation of the prediction system.

User Story	Development Tasks
Enter and update land data	Design data fields, create input form, implement database storage
Validate land records	Define validation rules, implement validation, display error messages
Analyze historical data	Collect sample dataset, clean data, identify important attributes
Identify delay factors	Analyze historical records, identify delay indicators, prepare features for prediction

Actionable Development Tasks

1. Design the land acquisition database.
2. Create the land acquisition input form.
3. Implement data validation.
4. Store acquisition records in the database.
5. Prepare historical/sample datasets.
6. Clean and preprocess the dataset.
7. Identify important delay-related factors.
8. Prepare the dataset for the prediction model.

6. Estimate Effort Using Story Points

Assign story points to each selected user story using an appropriate estimation technique (e.g., Fibonacci sequence or Planning Poker) and justify your estimates.

The **Fibonacci sequence (1, 2, 3, 5, 8, 13)** is used for estimation. Higher points indicate greater complexity, uncertainty, and development effort.

User Story	Story Points	Justification
Enter and update land data	5	Requires UI, database, and CRUD operations
Validate land records	3	Requires validation rules and error handling
Analyze historical data	5	Requires data collection, cleaning, and analysis
Identify delay factors	5	Requires analysis of multiple acquisition parameters
Predict delay risk	8	Involves developing and integrating a prediction model
Generate early warnings	3	Requires risk-level checking and notification logic
Dashboard	5	Requires data visualization and integration
Reports	3	Requires report generation and formatting

Total Estimated Story Points

37 story points

For the **first sprint**, the team can select approximately **13–18 points**, depending on its actual capacity.

7. Present the Sprint Goal and Expected Deliverables

Define a clear sprint goal and describe the expected sprint deliverables, including the features to be completed at the end of the sprint.

Sprint Goal

“To establish a reliable land acquisition data management and analysis foundation that prepares the system for early prediction of potential acquisition delays.”

Expected Sprint Deliverables

At the end of Sprint 1, the team is expected to deliver:

- 1. Land Acquisition Data Entry Module**
 - Add and update land acquisition records.
 - Store data securely in the database.
- 2. Data Validation**
 - Validate required fields.
 - Detect incomplete or invalid information.
- 3. Historical Data Preparation**
 - Collect and organize sample/historical acquisition data.
 - Clean and preprocess the dataset.
- 4. Delay Factor Analysis**
 - Identify factors such as:

- Documentation issues
- Landowner response
- Compensation status
- Legal disputes
- Government approvals
- Survey/measurement issues

5. Prediction Preparation

- Prepare relevant features and datasets for the delay prediction model.

Final Expected Outcome

The completed project will provide a **predictive analytics dashboard** that can analyze land acquisition information, estimate the likelihood of delays, identify important delay factors, and provide **early warnings** to officials so that corrective action can be taken before delays significantly affect infrastructure projects.

Rubrics for Calculation

Criteria	Excellent (5 Marks)	Good (3 Marks)	Satisfactory (2 Marks)	Needs Improvement (1 Mark)
Project Vision & Stakeholder Identification (30%)	Project vision is clear, comprehensive, and aligned with the problem statement. All relevant stakeholders and their roles are correctly identified.	Vision is clear with minor omissions. Most stakeholders are correctly identified.	Vision is partially defined. Some stakeholders are missing or incorrectly identified.	Vision is unclear or incomplete. Stakeholders are poorly identified or missing.
Epics & User Stories (25%)	Epics are well-defined and user stories are complete, follow Agile format, and fully capture functional requirements.	Epics and user stories are mostly complete with minor formatting or requirement issues.	User stories are partially complete with limited Agile structure.	User stories are incomplete, unclear, or do not follow Agile practices.
Acceptance Criteria (20%)	Acceptance criteria are specific, measurable, testable, and aligned with all user stories.	Acceptance criteria are mostly clear and testable with minor gaps.	Acceptance criteria are partially defined and lack clarity or completeness.	Acceptance criteria are missing or not aligned with user stories.
Product Backlog Prioritization & Sprint Planning (15%)	Product backlog is logically prioritized, sprint backlog is realistic, and task breakdown is complete.	Backlog prioritization and sprint planning are mostly appropriate with minor issues.	Prioritization or sprint planning is partially complete with limited justification.	Backlog prioritization and sprint planning are poorly organized or incomplete.
Story Point Estimation, Sprint Goal & Presentation (10%)	Story point estimation is well justified, sprint goal is clear, deliverables are realistic, and presentation is professional and well organized.	Estimation and sprint goal are mostly appropriate with minor inconsistencies. Presentation is clear.	Estimation or sprint goal lacks justification. Presentation is adequate but incomplete.	Estimation is unrealistic or missing. Sprint goal, deliverables, and presentation are unclear or incomplete.

Criteria	Mark
Project Vision & Stakeholder Identification (30%)	/5
Epics & User Stories (25%)	/5
Acceptance Criteria (20%)	/5
Product Backlog Prioritization & Sprint Planning (15%)	/5
Story Point Estimation, Sprint Goal & Presentation (10%)	/5
TOTAL	/25